

Prompt Engineering Assignment

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Q1.

Answer: Role: You are an AI tutor.

Prompt: Explain Artificial Intelligence, Machine Learning, Deep Learning, and Generative AI as connected concepts. Start with AI and then explain how Machine Learning, Deep Learning, and Generative AI fit inside it. Use one real-world example for each concept. Keep the explanation simple and suitable for a first-year college student. Present the answer with clear headings.

Q2.

Answer: Explain how an AI model understands a user's prompt. Describe the process of tokenization, understanding the context, and predicting the next token. Use a simple real-life analogy that a non-technical person can easily understand. Keep the explanation short and easy to follow.

Q3.

Answer: Explain why an AI model does not simply search the internet and copy answers. Describe how a Large Language Model generates responses using its training data and how probability helps it predict the next word. Use simple language with an example.

Q4.

Answer: Explain what a context window is in an AI model. Describe why it has a limit and what happens when a conversation becomes longer than the context window. Include one simple numerical example to make the explanation easy to understand.

Q5.

Answer: Create a two-column table comparing Training and Inference in AI. Explain what happens during each stage, when each stage takes place, and why both are important. Use simple words that are easy for beginners to understand.

Q6.

Answer: Poor Prompt: What is Machine Learning?

Improved Prompt: You are a computer science teacher. Explain Machine Learning to a first-year engineering student using simple language. Include one real-world example and explain where Machine Learning is used in daily life. Keep the answer within 200 words.

Why is the second prompt better?

The second prompt clearly defines the role, audience, and expected output, so the AI can generate a more useful and focused response.

Q7.

Answer: Role: You are a cloud computing expert.

Task: Explain Cloud Computing.

Context: The explanation is for a small business owner with no technical knowledge.

Instructions: Explain the basic idea, advantages, and common uses with simple examples.

Constraints: Avoid technical terms and keep the answer under 250 words.

Output Format: Use headings and bullet points.

Q8.

Answer: You are a professional business writer. Write a polite email to my manager requesting one day's leave due to a family function. The message should be professional, respectful, and limited to about 120 words.

Q9.

Answer: Act as an experienced HR Manager and draft a short policy note explaining the company's work-from-home guidelines for employees.

Q10.

Answer: I am a final-year engineering student preparing for campus placements. I have completed projects in Python and Machine Learning but have no work experience. Help me answer the interview question, "Tell me about yourself," in a confident and professional way that highlights my strengths.

Q11.

Answer: Explain Blockchain Technology in **exactly 80 words**. Do not use any technical terms or difficult words. Make the explanation easy enough for a school student to understand.

Q12.

Answer: Role: [ROLE], **Task:** [TASK], **Context:** [CONTEXT], **Instructions:** [INSTRUCTIONS]
Constraints: [CONSTRAINTS], **Output Format:** [FORMAT]

Use this template to explain any technical topic to any audience by replacing the placeholders with the required information.

Q13.

Answer: Classify the following customer review as **Positive**, **Negative**, or **Neutral**. Do not explain your answer.

Review:

"The delivery was fast, but the packaging was damaged."

Q14.

Answer: Example:

Customer Feedback: *"The support team solved my issue quickly."*

Category: **Compliment**

Now classify the following feedback into **Complaint** or **Compliment**.

Customer Feedback: *"I had to wait for three days before anyone replied to my email."*

Q15.

Answer: Classify each sentence as **Formal** or **Informal**.

Examples: Sentence: "Kindly submit the report by tomorrow."

Answer: Formal

Sentence: "See you later!"

Answer: Informal

Sentence: "Please attend the meeting at 10 AM."

Answer: Formal

Now classify these:

1. "What's up?"
2. "Thank you for your valuable feedback."

Q16.

Answer: Summarize the following text in five simple points.

""Artificial Intelligence is changing many industries by improving automation, decision-making, and customer service. It is widely used in healthcare, education, finance, and transportation. ""

Why do delimiters help?

They clearly separate the instructions from the text, making it easier for the AI to understand what needs to be summarized.

Q17.

Answer: Poor Prompt:

Explain databases.

Improved Prompt: Explain databases in simple language for a first-year college student. Include one real-life example.

Final Prompt:

You are a computer science teacher. Explain databases in simple English for a beginner. Keep the answer within 150 words and organize it under the headings **Definition**, **Example**, and **Benefits**.

Q18.

Answer: You are an experienced Career Counsellor.

I am a final-year B.Sc. Computer Science student who enjoys Python programming, data analysis, and problem-solving. I want a career with good growth opportunities.

Recommend three suitable career paths. Explain why each career matches my interests and mention the skills I should improve.

Q19.

Answer: A product costs ₹3,500. A shop gives a 15% discount and then adds 5% GST on the discounted price. Solve the problem step by step, showing each calculation clearly before giving the final amount.

Q20.

Answer: A car travels 240 km at a speed of 60 km/h.

Solve this problem using two different methods. Compare both methods and choose the most reliable final answer. Explain why the selected answer is correct.

Q21.

Answer: First, explain the general reasons why two products with similar features can have different prices. After explaining the reasons, apply them to compare two smartphones with similar specifications but different prices. Use simple language and include practical examples.

Q22.

Answer: Prompt 1: Identify the target audience for the new bakery.

Prompt 2: Suggest the best social media platforms to promote the bakery.

Prompt 3: Create a one-month content plan with post ideas.

Prompt 4: Suggest ways to measure the success of the marketing campaign.

Q23.

Answer: Prompt 1: Read the article and extract the main points.

Prompt 2: Use the extracted points to write a short summary in about 150 words.

Prompt 3: Based on the summary, create a 5-slide presentation outline with suitable slide titles.

Q24.

Answer: Explain the advantages and disadvantages of Artificial Intelligence. After writing the answer, review it for any factual mistakes, missing points, or unclear explanations. Then provide an improved version with the necessary corrections.

Q25.

Answer: Write a short apology email to a client for a delayed delivery. After writing the email, review it for tone, grammar, and clarity. If any improvements are needed, rewrite the email to make it more professional and polite.

Q26.

Answer: You are a professional content writer.

Write a LinkedIn post of about **300 words** announcing a new AI Training Program at Cambridge Infotech. The post should target working professionals, highlight the benefits of learning AI, and encourage readers to enroll. Use a professional and engaging tone.

Q27.

Answer: Read the given research article and summarize it into **exactly five bullet points**. Include only the main findings and ignore the background information. Keep each point short and clear.

Q28.

Answer: Classify the following customer support messages into **Complaint, Inquiry, or Feedback**. Return only the category for each message.

1. "My order arrived two days late."
2. "Can you tell me your business hours?"
3. "The website is very easy to use."
4. "I received a damaged product."
5. "Do you offer free shipping?"

Q29.

Answer:

Extract the following details from the candidate profile:

- Name
- Job Title
- Years of Experience
- Key Skills

Candidate Profile:

"Rahul Sharma is a Data Analyst with 4 years of experience. He has strong skills in Python, SQL, Excel, and Power BI."

Return the information in a clear list.

Q30.

Answer: Translate the following paragraph into **Hindi**. Keep technical terms unchanged and use simple, natural language.

"Renewable energy comes from natural sources like sunlight, wind, and water. It helps reduce pollution and protects the environment for future generations."

Q31.

Answer: Rewrite the following message in three different tones: Professional, Friendly, and Formal.

Message: "hey can u pls send the report by tmrw eod, thanks"

Q32.

Answer: Read the following information and answer the question using only the given context.

Context:

Electric vehicles use rechargeable batteries instead of petrol or diesel. They produce less pollution and help reduce carbon emissions. Many governments encourage people to use electric vehicles through subsidies.

Question:

Why are electric vehicles considered environmentally friendly?

Q33.

Answer: Explain the advantages of remote work in **exactly 6 bullet points**. Do not include any introduction or conclusion.

Q34.

Answer: Create a table comparing **Machine Learning** and **Deep Learning**.

Include these columns:

- Definition
- Data Requirement
- Typical Use Case

Use simple language.

Q35.

Answer: Extract the following information from the resume and return it in JSON format.

Resume:

"Anita is a Software Engineer with 5 years of experience. She knows Python, Java, SQL, and Git."

Schema:

- name
- role
- experience_years
- skills (array)

Q36.

Answer: Using the same candidate information, return the details in XML format with proper opening and closing tags.

Q37.

Answer: Extract product review details using this schema:

- product_name
- rating
- sentiment
- main_complaint

If any information is missing, return **null** instead of guessing.

Q38.

Answer: Read the customer complaint below.

Classify it as **Billing**, **Product**, or **Delivery**.

Identify the main issue, suggest a suitable solution, and return everything as a single JSON object.

Q39.

Answer: Convert the following business requirement into CSV-style output with sample records.

Requirement:

"Find all customers who spent more than ₹10,000 last month."

Include suitable column names.

Q40.

Answer: You are a Python developer.

Write a Python function that checks whether a number is prime.

Requirements:

- Handle negative numbers.
- Display an appropriate error message.
- Include example input and output.
- Write clean and simple code.

Q41.

Answer: Explain the following Python code line by line for a beginner. Describe what each line does and why it is used.

```
num = 5
```

```
fact = 1
```

```
for i in range(1, num + 1):
```

```
    fact *= i
```

```
print(fact)
```

Q42.

Answer: Debug the following Python code.

Code:

```
numbers = [10,20,30]
```

```
print(numbers[3])
```

Error:

IndexError: list index out of range

Expected

Output:

Display a valid list element without any error.

Explain the mistake and provide the corrected code.

Q43.

Answer: Optimize the following Python function to improve performance without changing the output.

Compare the original code and the optimized version. Explain the improvements made.

Q44.

Answer: Refactor the given Python code by:

- Creating proper functions.
- Using meaningful variable names.
- Following PEP 8 coding style.
- Keeping the output unchanged

Q45.

Answer: Generate unit test cases for a Python function that calculates the average of a list of numbers.

Include:

- 3 normal test cases
- 2 edge cases

- 2 invalid input cases

Use Python's unittest framework.

Q46.

Answer: Convert the following business requirement into an SQL query.

Requirement:

"List all employees whose department is Sales and salary is above ₹60,000."

Table

Name:

employees

Columns:

employee_id, name, department, salary

Q47.

Answer: You are a Senior Data Analyst.

Analyze the given dataset and describe:

- Number of rows
- Number of columns
- Data types
- Missing values
- Possible data quality issues

Present the results in a simple report.

Q48.

Answer: Analyze the dataset description.

Identify the columns with missing values and suggest the best method to handle each one. Explain why each method is suitable.

Q49.

Answer: Perform Exploratory Data Analysis on a sales dataset containing Date, Product, Region, and Revenue.

Identify:

- Important patterns
- Outliers
- Relationships between variables

Summarize your findings clearly.

Q50.

Answer: Suggest three suitable charts for the sales dataset.

For each chart, mention:

- Chart type
- X-axis
- Y-axis
- Business question answered

Q51.

Answer: A dataset has **Customer_Churn** as the target variable.

Suggest suitable Machine Learning algorithms.

Explain:

- Why each algorithm is appropriate
- Required preprocessing steps
- Evaluation metrics to use

Q52.

Answer: Analyze the business dataset from a business perspective.

Identify:

- Important trends
- Risks
- Opportunities
- Recommended actions

Focus on business insights instead of only describing the numbers.

Q53.

Answer: Create a reusable Data Analysis Prompt Template.

Role: [ROLE]

Dataset Description: [DATASET DESCRIPTION]

Business Objective: [BUSINESS OBJECTIVE]

Tasks:

- Understand the dataset
- Check missing values
- Suggest data cleaning methods
- Perform EDA
- Recommend visualizations
- Identify important patterns
- Generate business insights
- Suggest suitable Machine Learning models

Output

Format: Provide the results as a structured report with clear headings.